

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-AURA-001

Related Functional Requirement(s): FR-AURA-001

Objective: Implement a data processing pipeline to decode, normalize, and transform AURA input payloads into a canonical XML string.

Target Cluster Configuration:

- Cluster Name: AURA Processing Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2-5 x Standard_DS3_v2 (autoscaling)
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: xml, json, xslt

Source Data Details:

- I_AURA_INPUT_BASE64: Input payload encoded in Base64
- POLICY_NUMBER: Policy number associated with the input payload

Target Data Details:

- PARSED_AURA_OUTPUT: Canonical XML string
- POLICY_NUMBER: Policy number associated with the output

Job Flow / Pipeline Stages:

- Read I_AURA_INPUT_BASE64 and POLICY_NUMBER inputs
- Base64-decode I_AURA_INPUT_BASE64
- Progressive decoding loop to normalize AURA_OUTPUT
- XSLT transformation to produce canonical XML

Data Transformations / Business Logic:

- Step 1: Base64-decode I_AURA_INPUT_BASE64
- Step 2: Progressive decoding loop to normalize AURA_OUTPUT
- Step 3: XSLT transformation to produce canonical XML

Error Handling and Logging:

- Log exception message and current AURA_OUTPUT string for diagnostic purposes
- Increment error count

TR-FILE-001**Related Functional Requirement(s): FR-FILE-001**

Objective: Generate a 10K flat file with record-level fields derived through a mapplet and file header/filename derivations.

Target Cluster Configuration:

- Cluster Name: File Generation Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2-5 x Standard_DS3_v2 (autoscaling)
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: mplt_BNCPLS_IF23_10K_Field_Calc

Source Data Details:

- T_APPLICATION_STAGING: Application data
- T_BENEFICIARY: Beneficiary data
- \$\$M_SRC_SQL: SQL query to select eligible applications and associated beneficiaries/policies

Target Data Details:

- File_Structure_10K: 10K flat file target

Job Flow / Pipeline Stages:

- Use \$\$M_SRC_SQL to select eligible applications and associated beneficiaries/policies
- Join application data with T_BENEFICIARY data
- Derive Field_1..Field_n using mplt_BNCPLS_IF23_10K_Field_Calc mapplet
- Derive File_Path_Name using EXP_Calc_10K_FileName expression
- Apply Transaction Control to split output into multiple files

Data Transformations / Business Logic:

- Step 1: Join application data with T_BENEFICIARY data
- Step 2: Derive Field_1..Field_n using mplt_BNCPLS_IF23_10K_Field_Calc mapplet
- Step 3: Derive File_Path_Name using EXP_Calc_10K_FileName expression

Error Handling and Logging:

- Log errors and exceptions during file generation

TR-FILE-002

Related Functional Requirement(s): FR-FILE-002

Objective: Split the output into multiple files, enforcing a maximum of approximately 10,000 records per physical file.

Target Cluster Configuration:

- Cluster Name: File Splitting Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2-5 x Standard_DS3_v2 (autoscaling)
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: Transaction Control

Source Data Details:

- Output from TR-FILE-001

Target Data Details:

- Multiple 10K flat files

Job Flow / Pipeline Stages:

- Use Transaction Control to group rows into transactions
- Commit and initiate a new output file context when record count reaches \$\$MAX_ROWS_PER_FILE
- Derive unique filename for each output file using EXP_Calc_10K_FileName

Data Transformations / Business Logic:

- Step 1: Use Transaction Control to group rows into transactions
- Step 2: Commit and initiate a new output file context when record count reaches \$\$MAX_ROWS_PER_FILE

Error Handling and Logging:

- Log errors and exceptions during file splitting

TR-MAP-001

Related Functional Requirement(s): FR-MAP-001

Objective: Derive Field_1..Field_n using the mplt_BNCPLS_IF23_10K_Field_Calc mapplet.

Target Cluster Configuration:

- Cluster Name: Mapplet Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2-5 x Standard_DS3_v2 (autoscaling)
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: mplt_BNCPLS_IF23_10K_Field_Calc

Source Data Details:

- Application and beneficiary/policy data from SQ_T_APPLICATION_STAGING

Target Data Details:

- Derived Field_1..Field_n

Job Flow / Pipeline Stages:

- Use mplt_BNCPLS_IF23_10K_Field_Calc to centralize business field derivations
- Apply field standardization, fixed-width/padded formatting, code translations, date normalization, and string cleansing as per 10K specification

Data Transformations / Business Logic:

- Step 1: Use mplt_BNCPLS_IF23_10K_Field_Calc to derive Field_1..Field_n

Error Handling and Logging:

- Log errors and exceptions during mapplet execution

TR-EXP-001

Related Functional Requirement(s): FR-EXP-001

Objective: Derive File_Path_Name using the EXP_Calc_10K_FileName expression.

Target Cluster Configuration:

- Cluster Name: Expression Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2-5 x Standard_DS3_v2 (autoscaling)
- Autoscaling: Enabled

- Auto Termination: 30 minutes
- Libraries Installed: EXP_Calc_10K_FileName

Source Data Details:

- Input data from upstream transformations

Target Data Details:

- Derived File_Path_Name

Job Flow / Pipeline Stages:

- Use EXP_Calc_10K_FileName to derive File_Path_Name based on interface code, business date/batch id, sequence, and extension
- Pass through Field_1..Field_n from maplet to Transaction Control step

Data Transformations / Business Logic:

- Step 1: Use EXP_Calc_10K_FileName to derive File_Path_Name

Error Handling and Logging:

- Log errors and exceptions during expression execution

TR-JAVA-001

Related Functional Requirement(s): FR-JAVA-001

Objective: Handle errors and logging in the Java transformation.

Target Cluster Configuration:

- Cluster Name: Java Transformation Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2-5 x Standard_DS3_v2 (autoscaling)
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: Java transformation libraries

Source Data Details:

- Exceptions during AURA payload parsing or XSLT transformation

Target Data Details:

- Logged error messages and exception details

Job Flow / Pipeline Stages:

- Log exception message and current AURA_OUTPUT string for diagnostic purposes
- Increment error count

Data Transformations / Business Logic:

- Step 1: Log exception message and current AURA_OUTPUT string

Error Handling and Logging:

- Log errors and exceptions during Java transformation

TR-PARAM-001**Related Functional Requirement(s): FR-PARAM-001****Objective: Configure parameters for the session/workflow.****Target Cluster Configuration:**

- Cluster Name: Parameter Configuration Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2-5 x Standard_DS3_v2 (autoscaling)
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: Parameter configuration libraries

Source Data Details:

- Parameters defined in the parameter file or workflow variable

Target Data Details:

- Configured parameters for the session/workflow

Job Flow / Pipeline Stages:

- Configure \$\$M_SRC_SQL, \$\$M_XSLT_FILE, \$\$AURA15_M_XSLT_FILE, \$\$OUT_DIR, \$\$FILE_PREFIX, \$\$FILE_EXT, \$\$BATCH_ID, \$\$RUN_ID, and \$\$MAX_ROWS_PER_FILE
- Ensure environment has read access to XSLT files and write permissions on \$\$OUT_DIR

Data Transformations / Business Logic:

- Step 1: Configure parameters for the session/workflow

Error Handling and Logging:

- Log errors and exceptions during parameter configuration